# Differentiated complete-owner control-shift naturality, covariance-normal force repair, and the strict-margin frontier

TECT verification-first repository
claim `A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION`

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## 1. Purpose and scope

R-127 ended with a proposed common source-Hessian identity. An authority audit changes the correct description of that successor. R-119 Theorem 5.1 already proves the complete fixed-chart *control-shift* Hessian factorisation

$$D_h^2\mathcal{U}_{J,\pi}(h) = L_\pi^* B_{J,h} L_\pi, \tag{1.1}$$

including selfadjointness, vertical basicness, and quotient descent. R-104 already proves the exact once-owned scalar endpoint identity on every fixed temporally faithful chart. The present result does not register (1.1) a second time. It supplies the missing link and repairs the explicit force:

1. differentiate the recombined R-104 owner equality once and twice and combine it with the legal R-127 predictable adjoint;
2. distinguish this control variation from Gaussian/Malliavin source differentiation, which carries feedback derivatives;
3. correct the naked R-126 trace-excess force to the complete R-125 covariance-normal force;
4. state the exact refinement conjugacy and the source-block condition under which a legal reverse is genuinely the forward adjoint;
5. exhibit a conditional strict-margin rebudgeting route and forbid double spending between force completion and direct Hessian payment.

The scope is fixed finite cutoff, fixed positive rational floor, one R-104 temporally faithful chart, bounded smooth cylindrical-simple predictable controls, a fixed Gaussian base law and filtration, predictable fresh-root independent heat wherever R-104 disintegration is used, exhaustive orthogonal output clusters, and once-owned covariance packets. All differentiation is on the fixed cylindrical core. No cutoff-, chart-, control-, revisit-, or refinement-uniform production form estimate is claimed.

## 2. Differentiated complete-owner control pullback

Let

$$\mathcal{S}_\pi^{\mathrm{pred}} = \bigoplus_{b=1}^{B} L^2(\Omega, \mathcal{F}_{t_{b-1}}; \mathcal{S}_b), \qquad L_\pi h = \sum_b S_b h_b, \qquad S_b = (\Delta C_b)^{1/2}, \tag{2.1}$$

and

$$Z_h = S_\pi \xi + L_\pi h. \tag{2.2}$$

Put $\mathcal{E}_J := V_J^{\mathrm{ren}}$ and $\mathcal{E}_{J,\pi}(h) := \mathbb{E}\mathcal{E}_J(Z_h)$. This is the recombined R-104 renormalized-interaction endpoint, not yet the R-093 sextic stabilizer or source-cost square. R-104 gives, on the same fixed chart, an exact scalar identity

$$\mathcal{E}_{J,\pi}(h) = C_{J,\pi} + \mathcal{O}_{\mathrm{low/inj}}(h) + \sum_{\alpha} \mathcal{O}_\alpha(h), \tag{2.3}$$

where $\alpha$ ranges only over the active top-level R-104 owner summands. Root, output, row, and heat labels decorate those summands; forest and Schur coordinates are internal alternative representations and are not additional $\alpha$-terms. The constant contains only the fixed baseline. Every owner in (2.3) has the R-104 multiplicity.

### Theorem 2.1 (differentiated complete-owner pullback)

On the declared cylindrical core, (2.3) may be differentiated once and twice. Its legal source gradient is

$$\boxed{(\nabla_h \mathcal{E}_{J,\pi}(h))_b = \mathbb{E}\big[S_b^* D\mathcal{E}_J(Z_h) \mid \mathcal{F}_{t_{b-1}}\big]\,.} \tag{2.4}$$

If

$$g^{\mathrm{adm}}_{\mathrm{low/inj}} := \nabla_h \mathcal{O}_{\mathrm{low/inj}}, \qquad g^{\mathrm{adm}}_{\alpha} := \nabla_h \mathcal{O}_\alpha, \tag{2.5}$$

then

$$\boxed{P_{\mathrm{pred}} L_\pi^* D\mathcal{E}_J(Z_h) = g^{\mathrm{adm}}_{\mathrm{low/inj}} + \sum_\alpha g^{\mathrm{adm}}_\alpha.} \tag{2.6}$$

The common owner-complete control Hessian is

$$\boxed{H_\pi(h) := P_{\mathrm{pred}} L_\pi^* D^2\mathcal{E}_J(Z_h) L_\pi P_{\mathrm{pred}} = H^{\mathrm{adm}}_{\mathrm{low/inj}} + \sum_\alpha H^{\mathrm{adm}}_\alpha.} \tag{2.7}$$

It is selfadjoint and

$$\mathrm{Ker}\, L_\pi \subseteq \mathrm{Ker}\, H_\pi(h). \tag{2.8}$$

**Proof.** At fixed cutoff and positive floor every displayed endpoint is a smooth finite-dimensional rational/polynomial function with integrable derivatives on the bounded cylindrical core. R-104 proves (2.3) for every $h$ in that core, rather than at one selected value. Differentiation under the fixed Gaussian expectation is therefore valid. For predictable $k_b$,

$$\begin{aligned} D_h\mathcal{E}_{J,\pi}(h)[k] &= \mathbb{E}\langle D\mathcal{E}_J(Z_h), L_\pi k\rangle \\ &= \sum_b \mathbb{E}\langle \mathbb{E}[S_b^* D\mathcal{E}_J(Z_h) \mid \mathcal{F}_{t_{b-1}}], k_b\rangle, \end{aligned} \tag{2.9}$$

which proves (2.4). Differentiating the scalar equality (2.3) gives (2.6). A second derivative gives (2.7). Symmetry follows either from the scalar Hessian or directly from the symmetry of $D^2\mathcal{E}_J$. If $L_\pi k = 0$, both slots in (2.7) vanish, proving (2.8). □

The theorem is the differentiated R-104 owner incidence, not a new proof of the R-119 abstract terminal Hessian. It also does not license an independently guessed force for one owner: the right side of (2.6) consists of derivatives of the exact scalar owner functionals.

### Corollary 2.2 (refinement conjugacy)

Let $\iota : \mathcal{S}_\pi^{\mathrm{pred}} \to \mathcal{S}_{\pi'}^{\mathrm{pred}}$ be a fixed bounded linear admissible representation-preserving refinement map satisfying

$$L_{\pi'}\iota = L_\pi, \qquad \mathcal{E}_{J,\pi} = \mathcal{E}_{J,\pi'} \circ \iota. \tag{2.10}$$

Then

$$\boxed{g_\pi(h) = \iota^* g_{\pi'}(\iota h), \qquad H_\pi(h) = \iota^* H_{\pi'}(\iota h)\iota.} \tag{2.11}$$

This is the exact derivative form of R-104 recombined-total invariance. Labelled owner derivatives need not agree. Equality of quotient norms requires the corresponding isometric identification of admissible source spaces; a refinement that admits new controls is a new chart.

## 3. Control-shift versus Malliavin-source firewall

The phrase "derivative with respect to source block" has two meanings. In the fixed-law control variation used by Theorem 2.1,

$$h \longmapsto h + tk, \qquad D_h Z_h[k] = L_\pi k. \tag{3.1}$$

Every triangular feedback function is held fixed as an element of the predictable control space. This is the derivative in R-127's legal adjoint.

For a Gaussian source translation, the later feedback functions move. If

$$Z_h(\xi) = \sum_b S_b\{\xi_b + h_b(\xi_{<b})\}, \tag{3.2}$$

then

$$D_{\xi_b} Z_h = S_b + \sum_{c>b} S_c D_b h_c. \tag{3.3}$$

Writing $J_h = I + D_\xi h$, the complete formulas are

$$D_\xi(F \circ Z_h) = J_h^* L_\pi^* DF(Z_h), \tag{3.4}$$

$$\begin{aligned} D_\xi^2(F \circ Z_h)[u, v] = {} & D^2 F(Z_h)[L_\pi J_h u, L_\pi J_h v] \\ & + DF(Z_h)[L_\pi D_\xi^2 h[u, v]]. \end{aligned} \tag{3.5}$$

The last line is the feedback-connection term. R-119 control-Hessian selfadjointness does not remove it.

### Counterfixture 3.1 (the two source derivatives differ)

Take two scalar blocks and the bounded smooth predictable feedback

$$h_2(\xi_1) = \alpha \tanh \xi_1, \qquad Z = \xi_1 + \alpha \tanh \xi_1 + \xi_2, \qquad F(z) = \frac{z^2}{2}. \tag{3.6}$$

At $\xi_1 = \xi_2 = 0$, the control-shift Hessian and Gaussian-source Hessian are respectively

$$H_{\mathrm{ctl}} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \qquad H_{\mathrm{Mall}} = \begin{pmatrix} (1+\alpha)^2 & 1+\alpha \\ 1+\alpha & 1 \end{pmatrix}. \tag{3.7}$$

At $\alpha = 1$,

$$H_{\mathrm{ctl}} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \qquad H_{\mathrm{Mall}} = \begin{pmatrix} 4 & 2 \\ 2 & 1 \end{pmatrix}. \tag{3.8}$$

Thus the two derivatives cannot be interchanged. This fires `NG-2026-07-30-A13-CONTROL-MALLIAVIN-DERIVATIVE-CONFLATION`. The connection term in (3.5) is independently nonzero in the bounded smooth fixture

$$\widetilde{h}_2(\xi_1) = \frac{\beta}{2} \tanh^2 \xi_1, \qquad \widetilde{F}(z) = z. \tag{3.9}$$

At the origin its control-shift Hessian is zero, while $D_{\xi_1}^2(\widetilde{F} \circ Z) = \beta$. Thus both the feedback Jacobian and feedback-connection mechanisms are tested inside the declared bounded cylindrical class. This is a coordinate no-go, not an A1 or Nelson counterexample.

# 4. Covariance-normal force and the missing future variance

Fix one root and one exhaustive output cluster and suppress their labels. With R-125 notation set

$$\Phi = \mathbb{E}_f J, \qquad R = J - \Phi, \qquad \mathcal{V} = \mathbb{E}_f \|R\|^2, \qquad \mathcal{T} = \Theta - \|\Phi\|^2. \tag{4.1}$$

The covariance-normal endpoint is

$$\boxed{\mathcal{E}^{\mathrm{CN}} = \frac{1}{2}(\|\Phi\|^2 - \Theta) + \frac{1}{2}\mathcal{V} = \frac{1}{2}(\mathcal{V} - \mathcal{T}) = \frac{1}{2}(\mathbb{E}_f \|J\|^2 - \Theta).} \tag{4.2}$$

R-126 differentiates $\mathcal{T}$. Equation (4.2) shows that its Euler force is not, by itself, the derivative of $\mathcal{E}^{\mathrm{CN}}$.

Write

$$\dot{J}_H = DJ[H], \qquad \ddot{J}_{H,K} = D^2 J[H,K]. \tag{4.3}$$

### Theorem 4.1 (future-variance and covariance-normal calculus)

The exact first and second future-variance derivatives are

$$D\mathcal{V}[H] = 2\,\mathrm{Re}\,\mathbb{E}_f \langle R, \dot{J}_H \rangle, \tag{4.4}$$

$$\boxed{\begin{aligned} D^2\mathcal{V}[H,K] = 2\,\mathrm{Re}\{&\mathbb{E}_f \langle \dot{J}_H, \dot{J}_K \rangle - \langle \mathbb{E}_f \dot{J}_H, \mathbb{E}_f \dot{J}_K \rangle \\ &+ \mathbb{E}_f \langle R, \ddot{J}_{H,K} \rangle\}. \end{aligned}} \tag{4.5}$$

In particular,

$$D^2\mathcal{V}[H,H] = 2\mathbb{E}_f \|\dot{J}_H - \mathbb{E}_f \dot{J}_H\|^2 + 2\,\mathrm{Re}\,\mathbb{E}_f \langle R, \ddot{J}_{H,H} \rangle. \tag{4.6}$$

The covariance-normal endpoint obeys

$$\boxed{D\mathcal{E}^{\mathrm{CN}}[H] = \mathrm{Re}\,\mathbb{E}_f \langle J, \dot{J}_H \rangle - \frac{1}{2} D\Theta[H] = \frac{1}{2}(D\mathcal{V} - D\mathcal{T})[H],} \tag{4.7}$$

$$\boxed{\begin{aligned} D^2\mathcal{E}^{\mathrm{CN}}[H,K] = {}& \mathrm{Re}\,\mathbb{E}_f \{\langle \dot{J}_H, \dot{J}_K \rangle + \langle J, \ddot{J}_{H,K} \rangle\} \\ &- \frac{1}{2} D^2\Theta[H,K] = \frac{1}{2}(D^2\mathcal{V} - D^2\mathcal{T})[H,K]. \end{aligned}} \tag{4.8}$$

**Proof.** Differentiate $\mathcal{V} = \mathbb{E}_f \|J\|^2 - \|\mathbb{E}_f J\|^2$ under the fixed conditional law. The first derivative gives (4.4); a second derivative gives the three terms in (4.5). Completing the centered square gives (4.6). Differentiate either of the equal expressions in (4.2) for (4.7)–(4.8). ☐

For the production current $J = c(W)^T \partial_i W$,

$$\dot{J}_H = (Dc\, H)^T \partial_i W + c^T \partial_i H, \tag{4.9}$$

$$\ddot{J}_{H,K} = (D^2 c[H,K])^T \partial_i W + (Dc\, H)^T \partial_i K + (Dc\, K)^T \partial_i H. \tag{4.10}$$

Every R-120 rational multiplier remains inside $c, Dc, D^2 c$, and the R-121 Cartan owner remains inside the skew part once.

Let $g^{\mathrm{adm}}_{\mathcal{V},r,m}$ and $g^{\mathrm{adm}}_{\mathcal{T},r,m}$ be the legal source pullbacks of the two derivatives after exhaustive owner assembly. The corrected production coordinate is

$$\boxed{g^{\mathrm{CN,adm}}_{r,m} = \frac{1}{2}\{g^{\mathrm{adm}}_{\mathcal{V},r,m} - g^{\mathrm{adm}}_{\mathcal{T},r,m}\}, \qquad H^{\mathrm{CN,adm}}_{r,m} = \frac{1}{2}\{H^{\mathrm{adm}}_{\mathcal{V},r,m} - H^{\mathrm{adm}}_{\mathcal{T},r,m}\}.} \tag{4.11}$$

For an affine terminal control incidence, the block formula is

$$(g^{\mathrm{CN,adm}}_{r,m})_b = \mathbb{E}\Big[S_b^* G^{\mathrm{CN}}_{r,m} \mid \mathcal{F}_{t_{b-1}}\Big]. \tag{4.12}$$

For a more general owner coordinate, $S_b$ is replaced by the explicitly proved *control-shift* incidence. It must not be replaced by the Malliavin derivative (3.3).

**Counterfixture 4.2 (naked trace-excess force is incomplete)**

Let $\eta$ be centered with variance one and put

$$J_z = z(1+\eta), \qquad \Phi_z = z, \qquad \Theta_z = z^2, \qquad \mathcal{V}_z = z^2. \tag{4.13}$$

Then

$$\mathbb{E}J_z^2 = 2z^2 = \Phi_z^2 + \mathcal{V}_z, \qquad \mathcal{T}_z = 0, \qquad \mathcal{E}_z^{\mathrm{CN}} = \frac{z^2}{2}. \tag{4.14}$$

Thus $D\mathcal{T} = D^2\mathcal{T} = 0$ while

$$D\mathcal{E}^{\mathrm{CN}}[H] = zH, \qquad D^2\mathcal{E}^{\mathrm{CN}}[H,K] = HK. \tag{4.15}$$

This fires `AUDIT-2026-07-30-A13-R126-COVARIANCE-NORMAL-FORCE-OMISSION`. The corrected half-difference (4.11) is exact. The fixture is an abstract conditional cluster, not a full A1 counterexample.

# 5. Complete endpoint, stabilizer, cost, and owner ledger

If "complete R-093 endpoint" means

$$G_J(Z) = V_J^{\mathrm{ren}}(Z) + \frac{3}{20}\int |Z|^6, \tag{5.1}$$

then the sextic covector is part of the endpoint derivative:

$$G_6(Z) = \frac{9}{10}|Z|^4 Z, \qquad (g_6)_b = \mathbb{E}[S_b^* G_6(Z) \mid \mathcal{F}_{t_{b-1}}]. \tag{5.2}$$

Its Hessian is the nonnegative form

$$D^2 S_6(Z)[H,K] = \frac{9}{10}\mathbb{E}\int \{|Z|^4 H\cdot K + 4|Z|^2 (Z\cdot H)(Z\cdot K)\}. \tag{5.3}$$

If instead the full source action is differentiated,

$$\mathcal{A}_J(h) = \mathbb{E}G_J(Z_h) + \frac{9}{20}\mathbb{E}\sum_b \|h_b\|^2, \tag{5.4}$$

then the source cost additionally contributes

$$\nabla\mathcal{A}_{\mathrm{cost}} = \frac{9}{10}h, \qquad D^2\mathcal{A}_{\mathrm{cost}} = \frac{9}{10}I. \tag{5.5}$$

R-127 equation (10.1) is complete only after the endpoint/action convention and these terms are declared.

The once-owned derivative ledger is:

| Object | Derivative ownership rule |
|---|---|
| $\Delta C_b$ | Once through $S_b = (\Delta C_b)^{1/2}$ in the legal source pullback. |
| $\Delta\Gamma_r$ | Once in the root trace; it is not identified with $\Delta C_b$. |
| Heat | Internal to the fixed or predictable conditional endpoint; no appended heat owner. |
| Output clusters | One exhaustive orthogonal resolution before recombination. |
| Mean and future variance | Once each through $\|\Phi\|^2 + \mathcal{V} = \mathbb{E}_f \|J\|^2$. |
| R-063 forest | Alternate coordinate for the covariance-normal endpoint; zero extra multiplicity. |
| Rational multipliers | Once inside $c, Dc, D^2c$, including spatial derivatives through order two. |
| Cartan | Once in the rational skew block, with recombined checksum 2680/729. |
| Low/injected/paid | Retain the distinct R-079/R-104 owners; no renaming of the three low objects. |
| Sextic | Once in the endpoint derivative (5.2)–(5.3). |
| Source cost | Once outside the endpoint derivative when the full action (5.4) is used. |

Appending the R-063 forest to the already reconstructed current-minus-trace endpoint duplicates it. Cancelling the R-121 2680/729 block from scalar exactness is likewise invalid; global exactness says the complete companion curl is present, not that the isolated rational Cartan coefficient vanishes.

## 6. Common terminal and legal reverse firewalls

The corrected fixed-chart control Hessian does not prove that an arbitrary root family is one R-079 common-terminal martingale. Let

$$\mathcal{F}_1 = \sigma(\xi_1), \qquad \mathcal{F}_2 = \sigma(\xi_1, \xi_2), \qquad \Phi_1 = \xi_1, \qquad \Phi_2 = 2\xi_1 + \xi_2. \tag{6.1}$$

Each coordinate has a legal predictable mean and innovation, but

$$\mathbb{E}[\Phi_2 \mid \mathcal{F}_1] = 2\xi_1 \neq \Phi_1. \tag{6.2}$$

The tower condition rules out one terminal $J$ with $\Phi_j = \mathbb{E}[J \mid \mathcal{F}_j]$. This fires `NG-2026-07-30-A13-ROOTWISE-COMMON-TERMINAL-INFERENCE`. It is a logical obstruction, not a production counterexample. The actual common-terminal and matching low/root trace partition must be proved before using the R-126 low/injected transport.

For temporal source projections $E_b$, the common Hessian blocks

$$H_{bc} = E_b H_\pi E_c \tag{6.3}$$

satisfy

$$H_{bc}^* = H_{cb}. \tag{6.4}$$

Thus a legal reverse is free once a production forward block is genuinely a two-sided compression of this common source Hessian.

This does not follow from a one-sided spatial output projection. With

$$H = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \tag{6.5}$$

one has

$$(PH)^* = HP \neq PH. \tag{6.6}$$

The valid two-sided blocks are $PH(I-P)$ and its adjoint $(I-P)HP$. This fires `NG-2026-07-30-A13-ONE-SIDED-SHELL-PROJECTION-ADJOINT`. For the R-126 probability-root $r$ and spatial-output $m$ labels, a successor must prove

$$A_{mr} = Q_m H_\pi Q_r \tag{6.7}$$

for one common source-space shell resolution, or an explicit intertwiner with the same consequence. Only then is $A_{rm} = A_{mr}^*$.

# 7. Corrected augmented Loewner target

Put every cross term in the single form convention

$$\operatorname{Re}\langle x, A_{\text{eff}} y\rangle. \tag{7.1}$$

If $T$ is one oriented Hessian half-block and the quadratic form contains $2\operatorname{Re}\langle x, Ty\rangle$, then $A_{\text{eff}} = 2T$. The factor cannot be guessed after estimating one orientation.

For source, sextic, and low channels define

$$E = 2\eta I - R, \qquad F = 2\zeta I - S, \qquad D = -L, \tag{7.2}$$

$$M_2 = \begin{pmatrix} E & -A_{\text{eff}}/2 \\ -A_{\text{eff}}^*/2 & F \end{pmatrix}, \qquad K = \begin{pmatrix} B \\ C \end{pmatrix}. \tag{7.3}$$

R-127's bounded-operator criterion says that complete domination is equivalent to

$$\boxed{D \succeq 0, \qquad \operatorname{Ran} K^* \subseteq \operatorname{Ran} D^{1/2}, \qquad M_2 - X_D^* X_D \succeq 0,} \tag{7.4}$$

where $D^{1/2} X_D = K^*$ is the Douglas reduced solution. A Moore–Penrose expression is permitted only after closed range is proved.

For one root/shell pair define the oriented blocks and their full cross-form operators by

$$T_{mr}^{\mathcal{V}} = Q_m H_{\mathcal{V}} Q_r, \quad T_{mr}^{\mathcal{T}} = Q_m H_{\mathcal{T}} Q_r, \quad T_{mr}^{\text{CN}} = \frac{T_{mr}^{\mathcal{V}} - T_{mr}^{\mathcal{T}}}{2}, \quad A_{\text{eff}}^{\bullet} = 2T_{mr}^{\bullet}. \tag{7.5}$$

The operator inserted into (7.3) must now be assembled from

$$H_{\text{CN}} = \frac{1}{2}(H_{\mathcal{V}} - H_{\mathcal{T}}), \tag{7.6}$$

plus the declared sextic/source-cost pieces. An R-126 bound for $H_{\mathcal{T}}$ does not bound (7.6). If $K_{\mathcal{V}}$ and $K_{\mathcal{T}}$ denote norms of the full cross-form operators $A_{\text{eff}}^{\mathcal{V}}$ and $A_{\text{eff}}^{\mathcal{T}}$, then without a signed cancellation the comparable same-channel majorant is only

$$K_{\text{CN}} \le \frac{K_{\mathcal{V}} + K_{\mathcal{T}}}{2}. \tag{7.7}$$

For oriented-block norms the factor $1/2$ is absent after conversion to the full $A_{\text{eff}}$ convention. No uniform production $K_{\mathcal{V}}$ is registered.

Conditionally on genuine production block estimates, R-127 gives

$$K_{\text{tri}} = \sqrt{8/7}\, C_{\text{mix}} 2^{-j_0 - 2C} + \sqrt{224/223}\, C_{\text{far}} 2^{-3j_0 - 4C}. \tag{7.8}$$

Shared channels add by triangle. Equation (7.8) is an acceptance test, not a production estimate for the corrected Hessian.

## 8. A conditional strict-margin rebudgeting route

The R-103 regular theorem holds for every positive pair of form allocations. Its displayed debts

$$\frac{1}{440}, \qquad \frac{3}{100} \tag{8.1}$$

are a conservative choice, not a rigid identity. Keep the fixed R-124 row cost $3/(125P)$, where $P = 4 + 10^{-12}$, and choose half of (8.1):

$$\eta_h = \frac{9}{20} - \frac{3}{125P} - \frac{1}{880}, \qquad \zeta_h = \frac{3}{20} - \frac{3}{200} = \frac{27}{200}. \tag{8.2}$$

The following computation is the scalar $R = S = 0$ specialization of (7.2)–(7.4), equivalently the case in which every adverse diagonal debt has already been absorbed into $\eta_h, \zeta_h$. In the general operator matrix, the required hypothesis is a lower bound for the full $M_2$, not a bound on $A_{\mathrm{eff}}$ alone. Under this specialization the corresponding two-channel threshold is

$$K_h = 4\sqrt{\eta_h \zeta_h} = 0.9780518670016728\ldots. \tag{8.3}$$

The old R-126 acceptance budget is

$$K_0 = 4\sqrt{\left(\frac{197}{440} - \frac{3}{125P}\right)\frac{3}{25}} = 0.9209323339075279\ldots. \tag{8.4}$$

If the *complete corrected non-low production form*, after the declared diagonal debts have been absorbed, were proved to satisfy $\|A_{\mathrm{eff}}\| \le K_0$, then (8.2) would leave the exact eigenmargin

$$\begin{aligned} \mu_h &= \eta_h + \zeta_h - \sqrt{(\eta_h - \zeta_h)^2 + K_0^2/4} \\ &= 0.02396011633137279\ldots > 0. \end{aligned} \tag{8.5}$$

An affine low vector $d$ could then be paid at cost at most $\|d\|^2/\mu_h$. A quadratic low block with $D \succeq d_0 I$ would fit if its reduced-solution loss is strictly below $\mu_h$.

The limiting scalar threshold as the flexible debts tend to zero is

$$4\sqrt{\left(\frac{9}{20} - \frac{3}{125P}\right)\frac{3}{20}} = 1.032279032045117\ldots. \tag{8.6}$$

Thus exact low cancellation is not arithmetically necessary. The route is strictly conditional: R-103's estimates are for the regular graph class and R-104 proves that their visitwise quantitative use does not automatically temporalise. Equations (8.2)–(8.6) do not supply the missing production operator or low norm.

## 9. Source-allocation firewall

At source coefficient $c = 9/20$, R-127's full completion is

$$\langle h, g\rangle + c\|h\|^2 = c\left\|h + \frac{g}{2c}\right\|^2 - \frac{1}{4c}\|g\|^2, \qquad \frac{1}{4c} = \frac{5}{9}. \tag{9.1}$$

It uses the whole displayed source square. To expose the allocation, for $0 < \lambda \le 1$ write

$$\boxed{\begin{aligned} \langle h, g\rangle + c\|h\|^2 = \lambda c\left\|h + \frac{g}{2\lambda c}\right\|^2 + (1-\lambda)c\|h\|^2 \\ - \frac{1}{4\lambda c}\|g\|^2. \end{aligned}} \tag{9.2}$$

At $\lambda = 1$, no source square remains for a second Hessian payment. Combining the 5/9 loss in (9.1) with the full direct-Hessian source budget is therefore a double spend. This fires `AUDIT-2026-07-30-A13-FORCE-COMPLETION-HESSIAN-DOUBLE-SPEND`.

The clean active route is the direct Taylor formula already registered in R-119:

$$\mathcal{E}(Z_0 + u) - \mathcal{E}(Z_0) = D\mathcal{E}(Z_0)[u] + \int_0^1 (1-t) D^2\mathcal{E}(Z_0 + tu)[u,u]\,\mathrm{d}t. \tag{9.3}$$

A Hessian bound $2\eta X + 2\zeta Y$ integrates to $\eta X + \zeta Y$. The low affine anchor is then paid once through the same augmented matrix. The force-square route remains an exact alternative if an explicit $\lambda$-split is declared.

## 10. Evidence map and smallest successor

The recorded route decisions are:

| Route | Verdict | Reusable conclusion |
|---|---|---|
| New common control Hessian | duplicate | R-119 already owns it; cite rather than recount. |
| Differentiate R-104 total | advanced | Gives exact legal owner covector/Hessian after recombination. |
| Control equals Malliavin derivative | failed | Feedback Jacobian and connection term survive. |
| Naked R-126 force | failed | Complete force is $(g_{\mathcal{V}} - g_{\mathcal{T}})/2$. |
| Append the forest | failed | Forest is an internal alternate coordinate, not a second force. |
| Rootwise common terminal | failed | Tower compatibility is independent and necessary. |
| Automatic one-sided reverse adjoint | failed; production intertwiner open | Use two-sided common-source blocks or prove an intertwiner. |
| Half-debt rebudgeting | advanced conditionally | Creates strict scalar margin if the production form meets the old budget. |
| Full force plus full Hessian payment | failed | The same source square would be spent twice. |
| Unified corrected Schur gap | live | Requires production variance, intertwiner, balanced, low, and anchor bounds. |

These decisions and the bounded-fixture, one-sided-verdict, diagonal-scope, cross-factor, and formal-promotion audits are registered as `EXP-000436-EXP-000450`.

The smallest noncyclic successor is

$$\boxed{\texttt{A13-CLASSII-PRODUCTION-AUGMENTED-SOURCE-HESSIAN- UNIFORM-SCHUR-GAP}}. \tag{10.1}$$

It must, uniformly in cutoff, floor, chart, control, homotopy time, visits, output resolution, and representation-preserving refinement:

1. identify the production root/shell blocks as two-sided compressions of the corrected common covariance-normal control Hessian;

2. prove the future-variance and trace-excess block constants in the same form convention, including balanced and Cartan channels;

3. establish (7.4) with a strict positive margin or a proved range cancellation, and bound the Douglas reduced solution;
4. prove the absolute low/injected anchor and the R-079 common-terminal, matching-trace bridge;
5. retain $\Delta C$, $\Delta\Gamma$, all six rows, heat, exhaustive clusters, R-063 forest, rational multipliers, 2680/729 Cartan, mean, sextic, and source cost exactly once.

Only after this theorem may the R-093 directed chart union be invoked toward $\mathrm{OVERLAP}_{\mathrm{src}}$ and the $q = 10/9$ Nelson bound.

## 11. Devil's-advocate review

1. **Objection:** R-128 republishes R-119's common Hessian. **DISMISSED.** Equation (1.1) is explicitly cited as prior authority. The new result is differentiated R-104 owner incidence and the covariance-normal correction.
2. **Objection:** differentiating an expectation identity could move the law or filtration. **VALID WITH SCOPE.** The theorem fixes both. A moving Gibbs law would add a score covariance and is outside Theorem 2.1.
3. **Objection:** source block means Gaussian source, so (2.4) misses feedback derivatives. **UPHELD against that reading.** Section 3 names (2.4) the control-shift derivative and gives bounded smooth exact fixtures for both the feedback Jacobian and connection term.
4. **Objection:** refinement conjugacy holds for any nonlinear map. **UPHELD against that claim.** Corollary 2.2 assumes a fixed bounded linear refinement and evaluates the refined objects at $\iota h$; otherwise a $D^2\iota$ chain-rule term appears.
5. **Objection:** R-126's $\mathcal{T}$-force already contains the forest, so no variance force is needed. **DISMISSED.** Equations (4.2) and (4.14)–(4.15) show an exact missing derivative.
6. **Objection:** the future-variance correction may be added and the R-063 forest appended as well. **UPHELD as duplication.** The forest is the alternate endpoint coordinate in R-125/R-104.
7. **Objection:** source Hessian selfadjointness proves the geometric reverse band. **UPHELD only after (6.7).** One-sided output projection fails by (6.5)–(6.6).
8. **Objection:** the half-debt arithmetic closes the low channel. **DISMISSED.** It supplies only a conditional scalar margin; it assumes $R = S = 0$, or that those diagonal debts were already absorbed. Production form and low reduced-solution bounds are missing.
9. **Objection:** the old $K_0$ bounds the corrected operator. **DISMISSED.** It is an acceptance budget. The missing $H_{\mathcal{V}}$ must be included in (7.6)–(7.7).
10. **Objection:** the 5/9 force loss and direct Hessian reserve are independent. **UPHELD as double spending.** Equation (9.2) displays the common source owner.
11. **Objection:** the executable pastes the new decimal margin. **DISMISSED.** Both routes load the R-103, R-124, and A1 upstream inputs and derive every threshold; decimals are labelled regression intervals.

## 12. Reproducibility

Primary exact audit:

```
E:\Dev\TECT.venv\Scripts\python.exe
  codes/foundations/
  a13_classii_owner_complete_source_pullback_
  covariance_normal_force_boundary.py
```

Non-importing independent audit:

```
E:\Dev\TECT.venv\Scripts\python.exe
  codes/foundations/
  a13_classii_owner_complete_source_pullback_
  covariance_normal_force_boundary_independent.py
```

Integrated authority, PDF, exploration, registry, and public-surface audit:

```
E:\Dev\TECT.venv\Scripts\python.exe
  codes/foundations/
  a13_classii_owner_complete_source_pullback_
  covariance_normal_force_boundary_verify.py
```

## 13. Result footer

```
Result ID: A13-CLASSII-OWNER-COMPLETE-SOURCE-PULLBACK-
  COVARIANCE-NORMAL-FORCE-BOUNDARY (R-128).
Precise statement: Theorem 2.1 and Corollary 2.2; derivative firewall in
  Section 3; Theorem 4.1 and Counterfixture 4.2; owner ledger in Section 5;
  reverse/source-block firewall in Section 6; conditional strict-margin and
  allocation firewalls in Sections 8-9.
Scope: fixed finite cutoff and positive floor; fixed R-104 chart, Gaussian
  law, and filtration; bounded smooth cylindrical-simple predictable core;
  predictable fresh-root-independent heat; exhaustive clusters and
  once-owned covariance where the R-125 bridge is used.
Dependencies: A1; R-079, R-093, R-103, R-104, R-119--R-127.
Evidence grade: ANALYTIC, EXACT, EXECUTED.
Reproduction command: E:\Dev\TECT.venv\Scripts\python.exe
  codes/foundations/a13_classii_owner_complete_source_pullback_
  covariance_normal_force_boundary_verify.py
Expected output: R-128 integrated PASS; primary and independent assertion
  counts, manifest, PDF, exploration, negative, and public-surface gates pass.
Falsification gate: failure of differentiated owner recombination, legal
  control adjoint, refinement conjugacy, control/Malliavin fixture,
  future-variance or covariance-normal derivatives, naked-force fixture,
  sextic/source-cost factors, tower or projection fixtures, rebudgeting,
  partial completion, authority/hash, PDF, exploration, registry, or
  public-surface check.
Tier before / after: T4 / T4; no promotion.
No-overclaim statement: no production root/shell intertwiner, corrected
  covariance-normal operator bound, balanced/low closure, absolute anchor,
  OVERLAP_src, Nelson, removals, interacting measure, Sector-A closure, or
  T5--T7 follows.
Next required action: prove A13-CLASSII-PRODUCTION-AUGMENTED-SOURCE-HESSIAN-
  UNIFORM-SCHUR-GAP for the corrected covariance-normal common control
  Hessian, with the common-terminal/matching-trace bridge and absolute low
  anchor included.
```